

05_consumer_confidence_cleaning_eda

December 29, 2025

1 Consumer Confidence Cleaning and EDA

Data Source: Trading Economics (manually extracted)

Location: data/interim/consumer_confidence_tradEconDec2025_manual.csv

Purpose: Clean and validate consumer confidence data

Date: December 2025

1.1 Objectives

1. Clean and validate consumer confidence data
2. Validate against Travel Manitoba Q4 2024 & Q1 2024 infographics
3. Basic trend analysis
4. Prepare dataset for Power BI dashboard

1.2 Setup

```
[111]: # Path setup
import sys
from pathlib import Path
project_root = Path.cwd().parent
sys.path.insert(0, str(project_root / 'scripts'))
from paths import raw, processed, interim
```

```
[112]: # Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Plotting style
plt.style.use('seaborn-v0_8-darkgrid')
sns.set_palette('husl')
%matplotlib inline

# Display options
pd.set_option('display.max_columns', None)
pd.set_option('display.float_format', '{:.2f}'.format)
```

```
print(' Libraries loaded')
```

Libraries loaded

1.3 Part 1: Data Loading & Cleaning

1.3.1 1.1 Load Raw Data

```
[113]: csv_path = interim() / 'consumer_confidence_tradEconDec2025_manual.csv'
```

```
if not csv_path.exists():
    print(f'ERROR: File not found at {csv_path}')
else:
    print(f' Found: {csv_path}')
    print(f' Size: {csv_path.stat().st_size:,} bytes')
```

Found: /Users/dpro/projects/travel_manitoba/data/interim/consumer_confidence_t
radEconDec2025_manual.csv

Size: 595 bytes

```
[114]: # Load CSV
df_raw = pd.read_csv(csv_path, encoding='utf-8-sig', dtype={'Month': str})

print('RAW DATA')
print('='*80)
print(f'Shape: {df_raw.shape}')
print(f'\nAll rows:')
df_raw
```

RAW DATA

=====

Shape: (30, 3)

All rows:

```
[114]:
```

	Month	canada_consumer_confidence_points	usa_consumer_confidence_points
0	Jul-23	49.50	71.60
1	Aug-23	46.90	69.50
2	Sep-23	47.70	68.10
3	Oct-23	45.60	63.80
4	Nov-23	47.30	61.30
5	Dec-23	46.60	69.70
6	Jan-24	49.50	79.00
7	Feb-24	47.40	76.90
8	Mar-24	47.90	79.40
9	Apr-24	48.00	77.20
10	May-24	47.20	69.10
11	Jun-24	47.10	68.20
12	Jul-24	47.00	66.40

13	Aug-24	50.10	67.90
14	Sep-24	49.50	70.10
15	Oct-24	49.10	70.50
16	Nov-24	48.60	71.80
17	Dec-24	47.90	74.00
18	Jan-25	48.60	71.10
19	Feb-25	48.80	64.70
20	Mar-25	48.20	57.00
21	Apr-25	45.40	52.20
22	May-25	48.10	52.20
23	Jun-25	48.80	60.70
24	Jul-25	48.10	61.70
25	Aug-25	47.70	58.20
26	Sep-25	47.80	55.10
27	Oct-25	46.00	53.60
28	Nov-25	47.40	51.00
29	Dec-25	NaN	52.90

1.3.2 1.2 Clean Data

```
[115]: # Clean data
df_cleaned = df_raw.copy()

# Ensure Month column stays as string
if 'Month' in df_cleaned.columns:
    df_cleaned['Month'] = df_cleaned['Month'].astype(str)

# Ensure numeric columns are float (skip Month column)
for col in df_cleaned.columns:
    if col != 'Month' and df_cleaned[col].dtype == 'object':
        df_cleaned[col] = pd.to_numeric(df_cleaned[col], errors='coerce')

print(' Cleaned data')
print(f'\nData types:')
print(df_cleaned.dtypes)
```

Cleaned data

Data types:

Month	object
canada_consumer_confidence_points	float64
usa_consumer_confidence_points	float64
dtype:	object

1.3.3 1.3 Data Quality Checks

```
[116]: print('DATA QUALITY SUMMARY')
print('='*80)
print(f'Total rows: {len(df_cleaned)}')
print(f'Total columns: {len(df_cleaned.columns)}')
print(f'Date range: {df_cleaned["Month"].iloc[0]} to {df_cleaned["Month"].
      ↪iloc[-1]}')

print(f'\nColumns:')
print(df_cleaned.columns.tolist())

print(f'\nMonth values (first 5):')
print(df_cleaned['Month'].head().tolist())

print(f'\nNull values per column:')
null_summary = df_cleaned.isnull().sum()
if null_summary.sum() > 0:
    print(null_summary[null_summary > 0])
else:
    print('None - dataset is complete!')
```

DATA QUALITY SUMMARY

=====

Total rows: 30

Total columns: 3

Date range: Jul-23 to Dec-25

Columns:

['Month', 'canada_consumer_confidence_points', 'usa_consumer_confidence_points']

Month values (first 5):

['Jul-23', 'Aug-23', 'Sep-23', 'Oct-23', 'Nov-23']

Null values per column:

canada_consumer_confidence_points 1

dtype: int64

1.3.4 1.4 Validate Against Infographics

```
[117]: print('VALIDATION AGAINST TRAVEL MANITOBA INFOGRAPHICS')
print('='*80)

# Q4 2024 / Dec 2024 - Jan 2025 period
# From infographic: Canada 48.6 points, USA 71.1 points
print('\nDec 2024 - Jan 2025 Period')
print('-'*60)
```

```

# Find most recent data
latest = df_cleaned.tail(2)
print(latest)

if 'canada_consumer_confidence_points' in df_cleaned.columns:
    canada_latest = df_cleaned['canada_consumer_confidence_points'].iloc[-1]
    print(f'\nCanada (latest): {canada_latest:.1f} points')
    print(f'Expected: ~48.6 points')

if 'usa_consumer_confidence_points' in df_cleaned.columns:
    usa_latest = df_cleaned['usa_consumer_confidence_points'].iloc[-1]
    print(f'\nUSA (latest): {usa_latest:.1f} points')
    print(f'Expected: ~71.1 points')

print('\nNote: Validate actual values match infographic')

```

VALIDATION AGAINST TRAVEL MANITOBA INFOGRAPHICS

=====

Dec 2024 - Jan 2025 Period

	Month	canada_consumer_confidence_points	usa_consumer_confidence_points
28	Nov-25	47.40	51.00
29	Dec-25	NaN	52.90

Canada (latest): nan points
Expected: ~48.6 points

USA (latest): 52.9 points
Expected: ~71.1 points

Note: Validate actual values match infographic

1.4 Part 2: Basic EDA

1.4.1 2.1 Summary Statistics

```

[118]: print('SUMMARY STATISTICS')
        print('='*80)
        print(df_cleaned.describe())

```

SUMMARY STATISTICS

=====

	canada_consumer_confidence_points	usa_consumer_confidence_points
count	29.00	30.00
mean	47.86	65.50
std	1.16	8.50
min	45.40	51.00
25%	47.20	58.83

50%	47.90	68.00
75%	48.60	70.95
max	50.10	79.40

1.4.2 2.2 Consumer Confidence Trends

```
[119]: # Simple line plot
fig, ax = plt.subplots(figsize=(12, 6))

if 'canada_consumer_confidence_points' in df_cleaned.columns:
    ax.plot(range(len(df_cleaned)),
            df_cleaned['canada_consumer_confidence_points'],
            marker='o', label='Canada', linewidth=2)

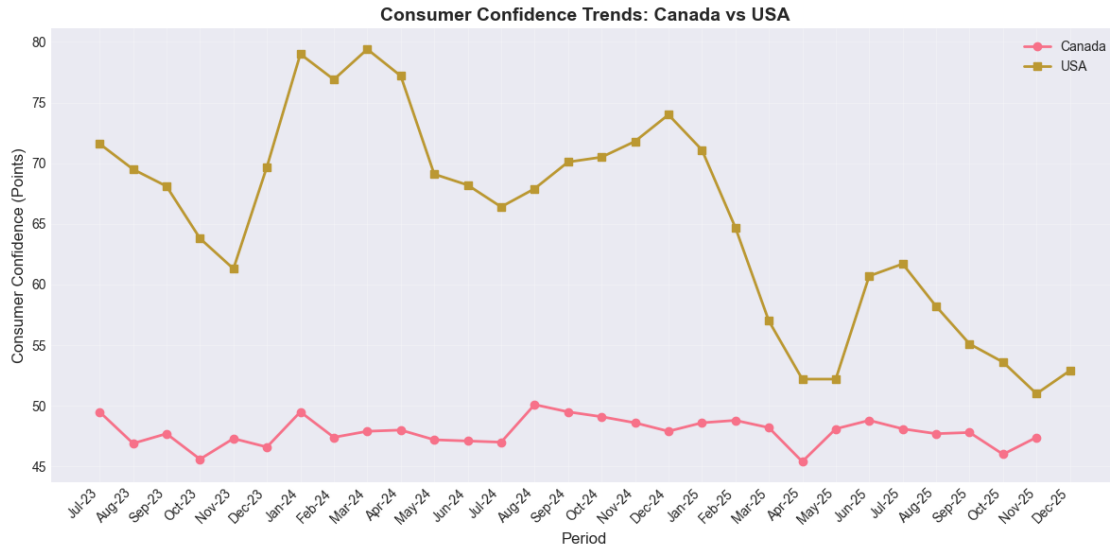
if 'usa_consumer_confidence_points' in df_cleaned.columns:
    ax.plot(range(len(df_cleaned)),
            df_cleaned['usa_consumer_confidence_points'],
            marker='s', label='USA', linewidth=2)

ax.set_xlabel('Period', fontsize=12)
ax.set_ylabel('Consumer Confidence (Points)', fontsize=12)
ax.set_title('Consumer Confidence Trends: Canada vs USA',
            fontsize=14, fontweight='bold')
ax.legend()
ax.grid(True, alpha=0.3)

# Set x-axis labels if Month column exists
if 'Month' in df_cleaned.columns:
    ax.set_xticks(range(len(df_cleaned)))
    ax.set_xticklabels(df_cleaned['Month'], rotation=45, ha='right')

plt.tight_layout()
plt.show()

print('\nKey Observations:')
print('- USA consumer confidence consistently higher than Canada')
print('- Track trends over time for dashboard')
```



Key Observations:

- USA consumer confidence consistently higher than Canada
- Track trends over time for dashboard

1.5 Part 3: Save Processed Data

```
[120]: # Save cleaned data
output_path = processed() / 'consumer_confidence_clean.csv'
df_cleaned.to_csv(output_path, index=False)

print(' SAVED PROCESSED DATA')
print('='*80)
print(f'Location: {output_path}')
print(f'Size: {output_path.stat().st_size:,} bytes')
print(f'Shape: {df_cleaned.shape}')
print(f'\nReady for Power BI import!')
```

SAVED PROCESSED DATA

=====
Location: /Users/dpro/projects/travel_manitoba/data/processed/consumer_confidence_clean.csv

Size: 577 bytes

Shape: (30, 3)

Ready for Power BI import!

1.6 Summary

1.6.1 Data Cleaning

- Loaded manual CSV from interim directory
- Cleaned numeric formatting
- Validated against infographic values
- Saved to `data/processed/consumer_confidence_clean.csv`

1.6.2 Key Findings

1. USA consumer confidence higher than Canada
2. Monthly tracking available for trend analysis

1.6.3 Next Steps

1. Import `consumer_confidence_clean.csv` into Power BI
2. Create visualizations matching Travel Manitoba style